

Mult_OS - LEFT manual

Version: 2026-09-18. Mult_OS has AS/Walk and TS modes. Hold DICE for about 3 seconds to change mode. A short unshifted DICE mutates AS or selects the next TS engine. SHIFT + DICE changes the mutation depth and mutates the scene; the four depths are Safe, Musical, Chaos and Wild.

Control map

Knob	AS normal layer	AS SHIFT layer
P1	Internal speed; under external clock, second MIDI layer transpose	Scale selection and voice-1 detune
P2	Base pitch	Pitch-walk range and voice-2 detune
P3	Harmonic spread and PM depth	Fold/tone and voice-3 detune
P4	Feedback and cross-modulation	Additional modulation depth
P5	Motion	Feedback damping
P6	Release	Space, stereo and motion

Knob	TS normal layer	TS SHIFT layer
P1	T1 pitch	Harmonic spread
P2	T2 pitch	Phase fold
P3	T3 pitch	Tone / brightness
P4	T1 decay	PM depth
P5	T2 decay	Feedback
P6	T3 decay	Motion and stereo

The three TS pads open three voices. In AS, individual pads accent PM, feedback and pitch-walk motion. A longer touch increases its influence.

CV and MIDI

CV1 is a trigger/clock input. CV2 works as continuous modulation until pulses are detected, then participates in clock/trigger handling. MIDI Note On takes pitch and triggers the voice group when CV does not own timing; held input notes are tracked and released. Start resets the walk step, while Stop clears incoming notes rather than persistently stopping AS. CC20-25 control the normal layer, CC26-31 the shifted layer, and CC74 controls the CV2 value. Local gates generate Note On/Off, panel movement generates CC, and local/CV timing generates Clock.

MIDI Clock uses 24 PPQN. CV has priority over MIDI, and an external timing source remains active for approximately 30 seconds. Start resets the walk step; Stop clears incoming notes.

MultOS is an Ellitone-inspired diceFriend OS built as a separate firmware family for Teensy 3.6 and Teensy 4.0 hardware.

Projects

- MultOS_36 - Teensy 3.6, analog stereo output.
- MultOS_40 - Teensy 4.0, PT8211 output.

Controls

- Short DICE: switch the current MultOS engine and apply mutation at the selected randomness depth.
- SHIFT + DICE: cycle randomness depth.
- Long DICE: toggle RS/walk mode and TS/touch mode.

- 4 engine LEDs: show the selected engine after switching.
- Randomness depth display: LED 0 = safe, LEDs 0-1 = musical, LEDs 0-2 = chaos.

Engines

- Engine 0: wave cluster, the most stable wavetable-like voice.
- Engine 1: glass FM, brighter and more metallic.
- Engine 2: chip swarm, sharper and more stepped.
- Engine 3: bounce, hollow/feedback-oriented and more generative.

Randomness Depths

- Safe: small drift, light detune, performance-friendly.
- Musical: voicing, scale, motion, and timbral mutation.
- Chaos: deeper pitch/feedback/crossmod/chaos changes plus deeper walk mutation.

Build

```
```sh cd MultOS_40 /Users/maciejjaciuk/.platformio/penv/bin/platformio run  
cd ../MultOS_36 /Users/maciejjaciuk/.platformio/penv/bin/platformio run ```
```